

Open science in stroke research

Measuring reproducibility and transparency, and what to do with the numbers

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Conflict-of-interest disclosure

Sebastian Fridman has nothing to disclose

- [x] No, nothing to disclose
- [ ] Yes, please specify

Company name	Honoraria / expenses	Consulting / advisory board	Funded research	Royalties / patent	Stock options	Ownership / equity position	Employee	Other (please specify)
None								

The congress's conflict-of-interest form, required as the second slide: the same two boxes and nine categories as the World Stroke Congress 2026 template.

01 Where I come from

In lesion neuroscience, the whole pipeline is open

I came to stroke from imaging and neuroscience. There, every step of a lesion study has open-source tools and shared data.

1 Segment

MRIcron 2007

ITK-SNAP 2006

LINDA 2016

nnU-Net 2021

DeepISLES 2025

2 Normalise

Cost-function masking 2001

Enantiomorphic 2008

SPM Clinical Toolbox 2012

ANTs 2008

3 Map the symptom

VLSM 2003

NiiStat

SVR-LSM 2014

LESYMAP 2018

4 Map disconnection

Lesion network mapping 2015

BCBtoolkit 2018

Lesion Quantification Toolkit 2021

5 Share

BIDS 2016

NeuroVault 2015

OpenNeuro 2021

Open stroke data

1,271
ATLAS v2.0 stroke MRIs with traced lesions; 955 public

250
ISLES 2022 acute MRI cases public under CC BY, of 400

230
people with aphasia, 902 sessions, CC0: Aphasia Recovery Cohort

1,715
admissions with clinical MRI, CC0: SOOP

Open data has not guaranteed open code: of 17 segmentation methods published on ATLAS v1.2, 12 reported no public code (Liew 2022).

02 Why open science

Different problems within the reproducibility crisis

Replication · Psychology

36%

of 100 published findings replicated with a significant result; effect sizes about half the originals. Open Science Collaboration, Science 2015

Statistical power · Neuroscience

21%

median statistical power of neuroscience studies. Button et al., Nature Reviews Neuroscience 2013

Confirmation · Preclinical cancer

6 of 53

landmark findings that Amgen could reproduce. Begley and Ellis, Nature 2012

Experience · Researcher survey

70%

of 1,576 researchers had failed to reproduce another group's experiment; 52% called it a significant crisis. Baker, Nature 2016

Why it happened

Small samples, flexible analyses, selective reporting. Four routine analysis choices raise a 5% false-positive rate to 61% (Simmons 2011); 63% of 2,155 psychologists admitted not reporting all their measures (John 2012).

Replication studies, power estimates and surveys answer different questions.

03 Where the fields next door are now

The rules now exist. Practice is catching up.

What they agreed on	
2013	Registered Reports launch at *Cortex*, a neuroscience and neuropsychology journal; more than 300 journals offer them now
2015	TOP guidelines for journals, 5,000+ signatories; revised as TOP 2025, which certifies, not just asks
2016	BIDS, one standard for organising and sharing neuroimaging data
2017	A manifesto for reproducible science; COBIDAS, the reporting standard for MRI studies
2024	*Psychological Science* makes open data and materials the default and checks that results reproduce
2025	PECANS, a Delphi consensus checklist for cognitive and neuropsychological studies, 206 experts

Where they are now

55%

of 274 social and behavioural science claims replicated; effects fell from r = 0.25 to 0.10. SCORE, *Nature* 2026

7% · 14%

of 2022 psychology articles were preregistered, or shared raw data; 3% and 2% in 2014-2017. Hardwicke 2024

1 of 62

Registered Reports at *Cortex*, 2018-2022, studied patients; sharing clinical data is the barrier named most. Binney 2025

Registered Reports: 44% positive results against 96% in standard papers (Scheel 2021). The best-known evidence that rigour alone restores replication (Protzko 2023) was retracted in 2024.

04 The clinical question

Different stroke trials can reach different conclusions

Preclinical	**1,026** neuroprotectants tested in animal stroke; none translated. Publication bias overstated efficacy by about a third. NXY-059: SAINT I positive, SAINT II negative.	O'Collins 2006; Sena 2010; Lees 2006; Shuaib 2007
Tirofiban	No benefit before thrombectomy in RESCUE BT (2022). Benefit without a large-vessel occlusion in RESCUE BT2 (2023). The trials enrolled different populations.	Qiu 2022, JAMA; Zi 2023, NEJM
Middle meningeal artery embolisation	Three trials in one NEJM issue: EMBOLISE positive (recurrence 4% vs 11%), STEM positive, MAGIC-MT neutral (6.7% vs 9.9%).	Davies, Fiorella, Liu 2024, NEJM

Open protocols, data and code help explain why results differ.

05 What open science is made of

Four questions a field should be able to answer about its own literature

Read it	Check it	Reuse it	Plan it
Can people read it?	**Can they check it?**	**Can they reuse it?**	**Was it planned in advance?**
Open access, and by which route	Data and code that someone else can run	Deposited, not promised on request	Registration and preregistration
Diamond, gold, hybrid, green, bronze or closed	Repository links, GitHub, data-availability statements	A repository or a controlled-access platform	Trial registries, PROSPERO, Registered Reports

> Reproducibility audits exist for neurology, cardiology and general medicine. **None existed for stroke.** Rauh 2020; Heckerman 2025; Scutt 2020

06 How we measured

A living pipeline measures the stroke literature every six months

Retrieve	OpenAlex, 2015 onward; stroke plus neurology, neuroscience, cardiology
Clean	Journal articles and reviews with a DOI; datasets, preprints, errata and chapters out
Detect	Open-access status, repository links, preprint versions, registration identifiers
Read	Data-availability statements in up to 1,500 open-access full texts per year, via Europe PMC, all four fields
Publish	Open code, open data, public dashboard; re-run every six months

140,281
stroke articles and reviews

16,362
stroke full texts read

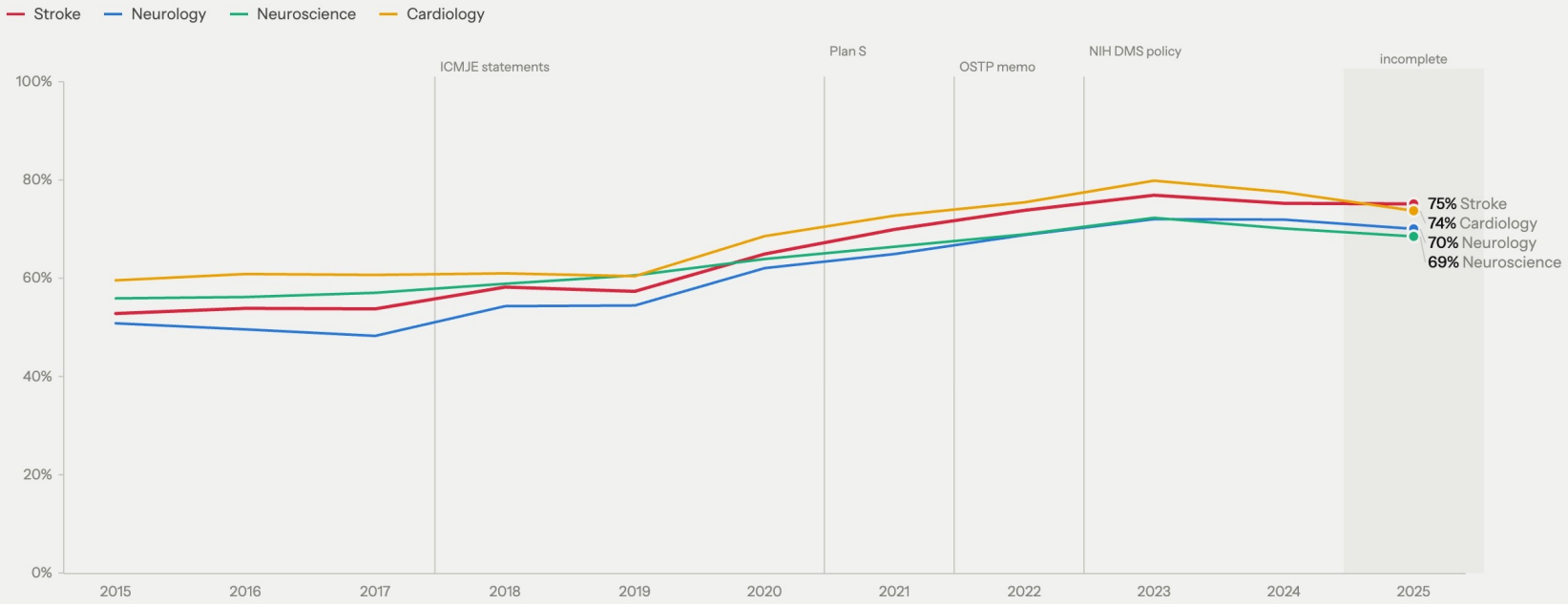
4
fields, 2015 to 2025



friseeb.github.io/stroke_open_science_outputs · Abstract and metadata signals are lower bounds; the full-text audit narrows the gap for data sharing.

07 Finding 1

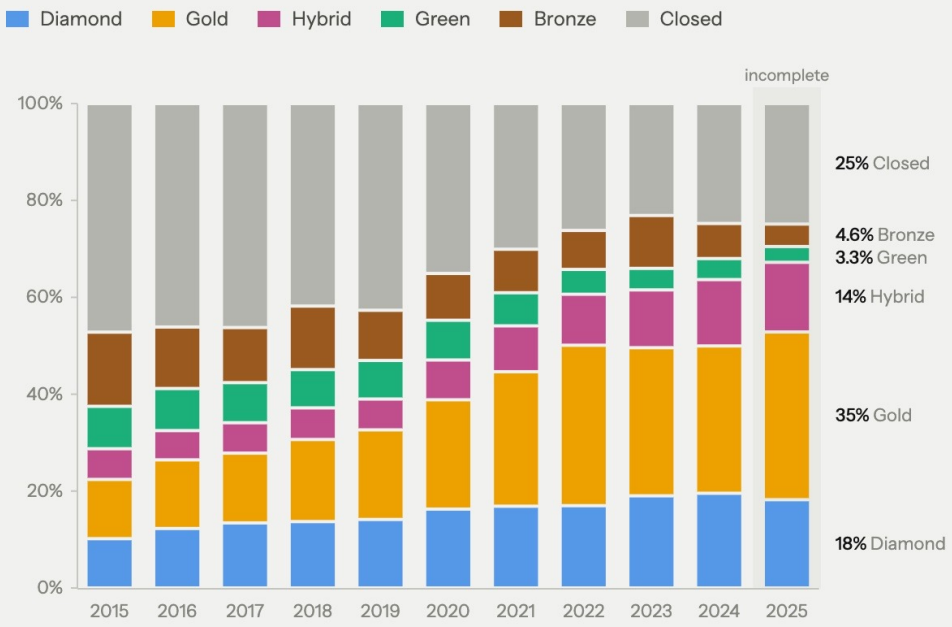
Stroke went open access: 53% in 2015, 75% in 2024



Open access = any OpenAlex status other than closed. Stroke 66% over the decade; cardiology 70%, neurology 62%, neuroscience 65% (its title term “neural” also catches machine-learning papers). 2025 is incomplete.

08 Finding 2

The growth in open access was paid for, not deposited



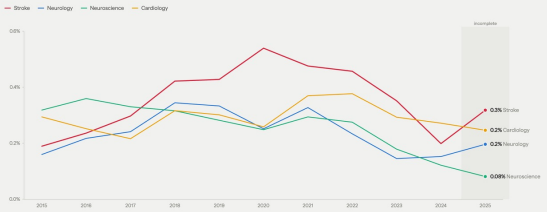
Route, 2015 to 2025	Share
Gold, paid journal	25%
Diamond, free journal	16%
Hybrid, paid in a subscription journal	10%
Bronze, free to read, no licence	10%
Green, repository copy	6%
Closed	34%

From 2015 to 2024 gold, a paid route, grew the most (+18 points); diamond (+9) outgrew hybrid (+7). Green fell from 9% in 2015 to 3% in 2025. Bronze can be withdrawn at any time.

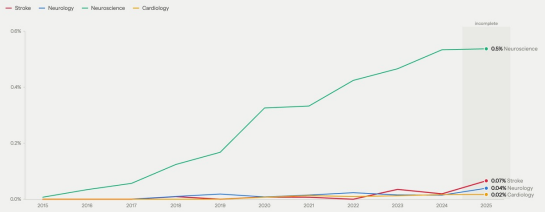
Open access became a transaction between funders and publishers. It did not come with openness of anything else.

09 Finding 3

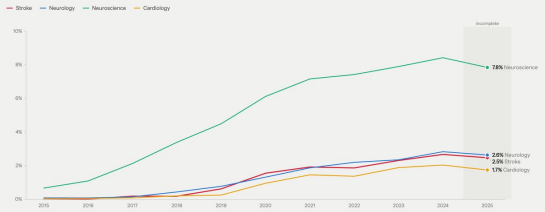
In abstracts, data and code did not move in ten years



Data in a public repository



GitHub or code link



Preprint version

Data in a repository

0.2%

of stroke articles in 2024; 0.2% in 2015, never above 0.6%

Code link

<0.1%

every year 2015-2024; GitHub in 2.5% of 2024 full texts

Preprint version

2.7%

in 2024; 1.5% in 2015-2025. A lower bound, partly publisher-driven

Registration ID in the abstract

5.3%

in 2024, from 2.8% in 2015; stroke ahead of cardiology, neurology and neuroscience

High impact is barely better at sharing: top-cited or IF 10+ articles (16% of the corpus) are 74% open access vs 65%, registration 10.0% vs 3.5%, but data 0.45% vs 0.34%, code 0.08% vs 0.06%. These are lower bounds from abstracts and metadata. So we read the full text.

10 What data-availability statements actually say

Write a statement, 2025

74%

from 14% in 2015; arrived with mandates

Say “available on request”, 2025

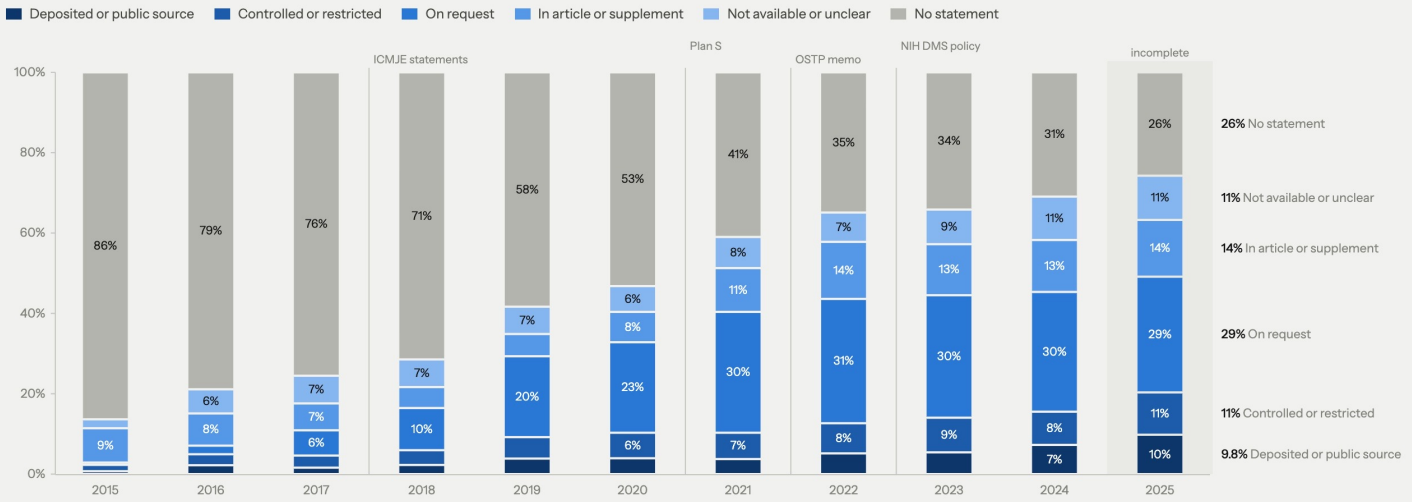
29%

from 0.6% in 2015; the largest answer

Deposit or name a public dataset, 2025

9.8%

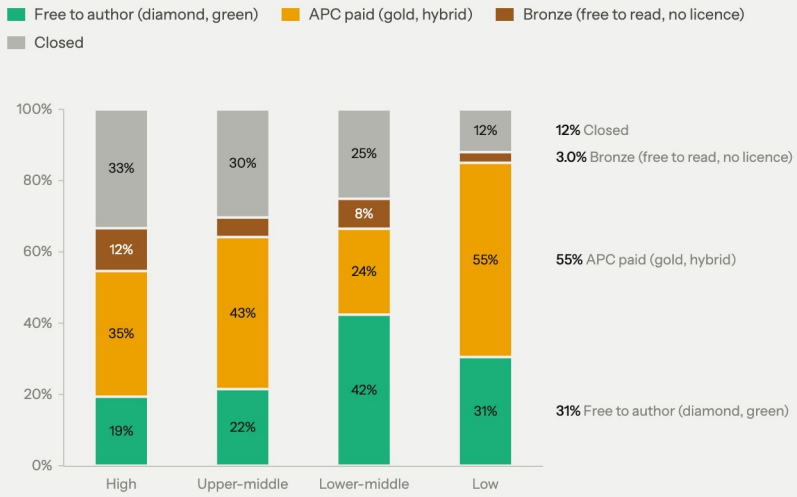
from 0.7% in 2015; own deposit 5.9%



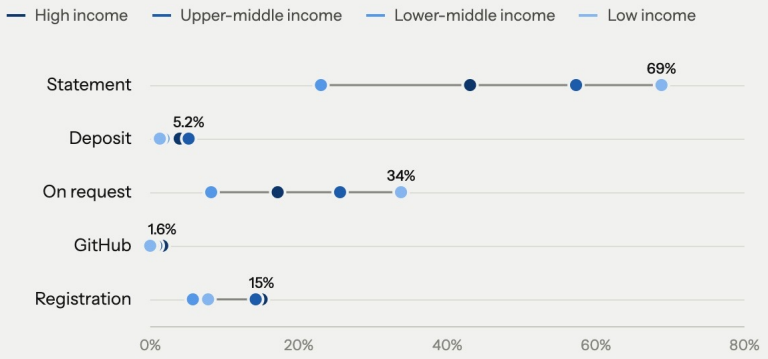
16,362 open-access stroke full texts, up to 1,500 per year, Europe PMC; optimistic for stroke overall. Elsewhere, “on request” delivers little: 6.8% of authors shared when asked (Gabelica 2022); 20% of 391 vascular and cognitive trials, median 241 days (Scutt 2020).

11 Who pays to be read, and who shares

Reading access has been solved for the global South by free routes. Reuse has not.



Open-access route by World Bank income group of the first author's country



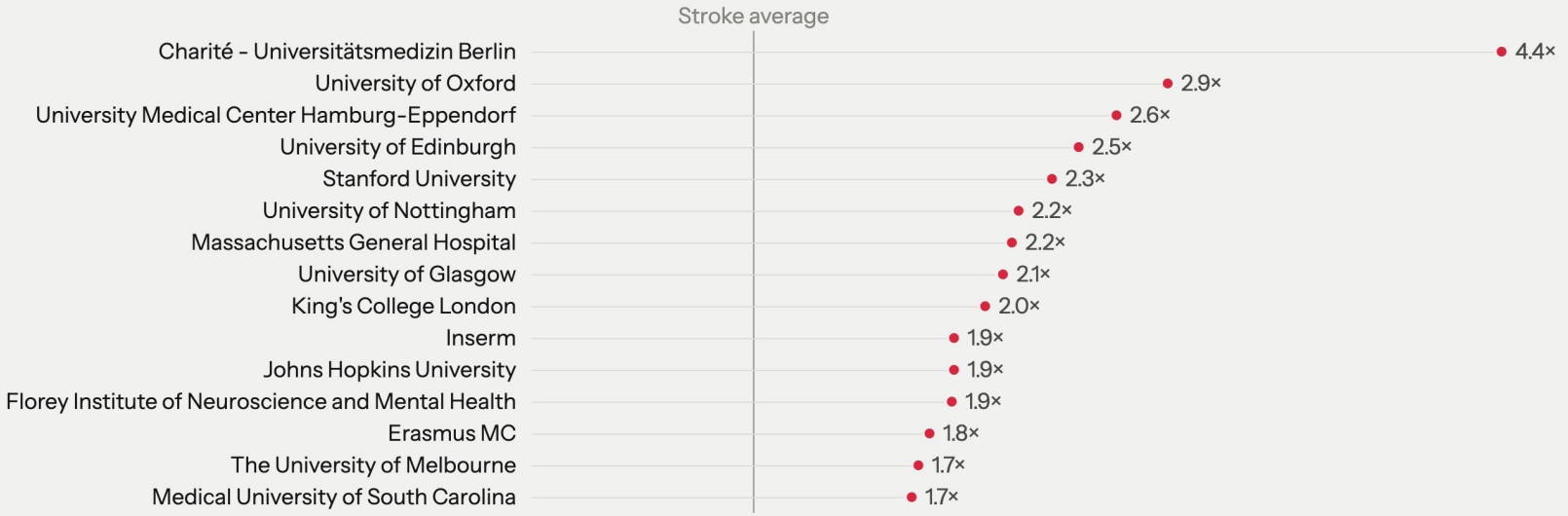
What the full text shows, by income group

Lower-middle-income authors publish 42% through routes that cost nothing (high income 19%), yet over 2015-2025 deposit or name public data in 1.8% (high 4.0%), mention GitHub in 0.3% (1.6%) and a trial registration in 5.7% (15.0%).

Income group known for 85% of the corpus: high 62%, upper-middle 32% (mostly China), lower-middle 6%, low 0.3% (400 articles). Korea: 5,490 first-author articles, 75% open access.

12 Who is already doing it

A commitment index, verified in the full text



Journals: Nature Communications, Brain Communications, PeerJ, NeuroImage Clinical lead; *Stroke* is about 15th of 99, at 1.5×. Funders: US federal institutes, ANR, Swiss Heart Foundation, Leducq, DFG. Countries: Belgium 2.4×, Germany 2.2×, Switzerland 2.0×, France 1.8×, UK 1.5×.

Index = mean, over data deposited or public source in the statement, GitHub and trial registration in the full text, of the entity's rate divided by the stroke audit rate (4.2%, 1.3%, 14.1%). 1 = stroke average. Institutions: the 60 with the most audited articles over the whole audit (each 82+ in 2015–2025); countries 50+, journals and funders 30+.

13 Why mandates alone did not work

Rules produced statements. Incentives produce data.

3.1% to 3.1%	Data obtainable from surgical trials before and after the ICMJE statement requirement.	Bergeat 2022, JAMA Netw Open
OR 0.9	Open-data badges in the only medical randomised trial, at BMJ Open. The same badges took Psychological Science from under 3% to 39% open data.	Rowhani-Farid 2020; Kidwell 2016
0 of 92	Biomedical promotion guidelines worldwide that mention data sharing, registration or open access.	Rice 2020, BMJ
44% vs 96%	Positive results in Registered Reports versus the standard literature. Changing when review happens changes what gets published.	Scheel 2021

Stroke researchers are not less honest than psychologists. They respond to the same incentives, and ours point the other way.

Cardiology survey, n = 198: funding named the most essential incentive; “policy alone will not lead to implementation” (Cobey 2024).

14 What to do next

A task force, a prize, and awareness

1

A WSO task force on open science

- A position statement: no stroke society has one
- Retire “available on request” in Stroke, IJS, ESJ and JSCVD: a repository, VISTA or Vivli, or a stated reason
- Registered Reports for stroke trials, with in-principle acceptance tied to WSO and ESO grants
- Embargoed sharing as the trial default, twelve months after publication

2

A prize for the teams that share

- The commitment index: verified in the full text, recomputed every six months
- Institutions, journals, funders and countries
- On the Einstein Foundation model, presented at this congress

3

Awareness

- The public benchmark, updated every six months
- Open-science items in promotion criteria: 0 of 92 guidelines today
- Targets to discuss: data in a repository or controlled-access platform for 25% of stroke articles by 2030; “on request” under 10%; one Registered Report track in a stroke journal by 2028

The statement, retiring “on request” and promotion criteria cost nothing. Registered Reports, the prize and the infrastructure need money, and the infrastructure mostly exists: VISTA, ATLAS, ENIGMA.

We measured ourselves. The numbers are not flattering, but they are measurable, which means they are improvable.

COLLABORATE

Stroke Open-Science Benchmark

friseeb.github.io/stroke_open_science_output

Explore it, challenge it, run it on your field

Code and data

github.com/Friseeb/stroke_open_science

Open source; the next run is January 2027

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Write to join the task force or to audit your own field

Disclosures: none.